

Ecommerce Business Analysis Using SQL

1. Introduction

Ecommerce companies generate large volumes of transactional and customer interaction data. Analyzing this data can help organizations improve customer experience, optimize marketing investments, identify operational inefficiencies, and drive business growth.

This project uses SQL to analyze ecommerce data and answer key business questions related to revenue performance, customer behavior, conversion rates, delivery operations, and growth opportunities.

2. Objectives

The main objectives of this project are:

- Identify underperforming product categories.
 - Evaluate the effectiveness of marketing spend.
 - Analyze website conversion performance.
 - Determine the most valuable customer segment.
 - Understand the impact of discounts on revenue.
 - Identify operational issues affecting customer experience.
 - Recommend KPIs for business growth.
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3. Dataset Overview

The dataset contains ecommerce transaction and business performance data with the following attributes:

- Order Date
- Revenue
- Orders
- Product Category
- Region

- Customer Segment
 - Marketing Spend
 - Website Visits
 - Discount Percent
 - Delivery Days
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4. Business Questions and Analysis

Question 1

Which product category is underperforming by region?

Analysis

Revenue and order volume were analyzed across product categories and regions.

Finding

Certain product categories generated significantly lower revenue compared to others within the same region.

Recommendation

Review pricing, inventory allocation, and customer demand for low-performing categories.

Question 2

Is higher marketing spend actually improving orders and revenue?

Analysis

Marketing spend was compared with revenue and order volume.

Finding

Higher marketing expenditure did not consistently result in higher revenue or more orders.

Recommendation

Evaluate marketing campaign efficiency and focus spending on higher-performing channels.

Question 3

Which regions are getting high website traffic but poor conversion rates?

Analysis

Conversion rate was calculated using orders and website visits.

Formula

Conversion Rate = Orders / Website Visits × 100

Finding

The West region received high website traffic but showed relatively weaker conversion performance.

Recommendation

Investigate customer journey, checkout experience, and regional marketing effectiveness.

Question 4

Which customer segment is most valuable?

Analysis

Revenue was aggregated by customer segment.

Finding

One customer segment contributed significantly more revenue than others.

Recommendation

Focus retention and loyalty strategies on high-value customer segments.

Question 5

Are discounts helping sales growth or hurting revenue quality?

Analysis

Average discount percentages were compared with revenue across categories and regions.

Finding

Higher discounts did not consistently translate into higher revenue.

Recommendation

Reduce unnecessary discounting and focus on improving value proposition.

Question 6

Which operational issue is impacting customer experience the most?

Analysis

Average delivery days were analyzed by region and compared against regional revenue.

Finding

Regions with longer delivery times generally generated lower revenue.

Conclusion

Delivery delays appear to be the most significant operational issue affecting customer experience.

Recommendation

Improve logistics efficiency and reduce delivery times.

Question 7

Which KPI should leadership monitor most closely to improve business growth?

Analysis

Monthly revenue trends were compared with discounts, delivery performance, website visits, and marketing spend.

Finding

Marketing spend showed the strongest relationship with business growth.

Recommended KPI

Marketing Spend Efficiency

Supporting KPIs:

- Monthly Revenue Growth

- Conversion Rate
 - Website Traffic
 - Average Delivery Days
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5. Key Insights

1. Some product categories underperform across regions.
 2. Marketing spend requires optimization to improve return on investment.
 3. High website traffic does not guarantee high conversion rates.
 4. Customer segments differ significantly in revenue contribution.
 5. Discounts alone are not driving revenue growth.
 6. Delivery delays negatively impact business performance.
 7. Marketing efficiency is a critical driver of growth.
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6. Business Recommendations

- Optimize marketing budget allocation.
 - Improve website conversion performance.
 - Reduce delivery times.
 - Focus on retaining high-value customers.
 - Minimize unnecessary discounting.
 - Monitor revenue growth and marketing efficiency regularly.
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7. Conclusion

This project demonstrates how SQL can be used to generate actionable business insights from ecommerce data. The analysis identified key growth opportunities, operational bottlenecks, and customer behavior patterns. The findings suggest that improving marketing efficiency and delivery performance can significantly contribute to sustainable business growth.